

ANUDEEP BURRA

Data Engineer | Python, SQL, ETL Pipelines, Cloud (AWS)

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UK Graduate Route Visa — Immediately Available, No Sponsorship Required

PROFESSIONAL SUMMARY

Data Engineer with 2+ years' professional IT experience and an MSc in Computing (Merit) from Sheffield Hallam University. Built and deployed a production-style **end-to-end ETL pipeline** in Python (Pandas, SQLAlchemy) that ingests live job market data via REST API, transforms and validates it, and loads it into a cloud PostgreSQL database (Supabase) on a fully automated daily schedule with CI/CD via GitHub. Delivered a Streamlit analytics dashboard processing **260,000+ records**. Strong background in SQL data validation and QA from enterprise banking/CRM systems at Infosys. Eligible to start immediately in the UK on the Graduate Route visa — **no sponsorship required**.

TECHNICAL SKILLS

Data Engineering: ETL/ELT pipeline design, REST API ingestion, data transformation & modelling, batch processing, cron/job scheduling, data validation

Languages: Python, SQL, Java, Bash

Python Libraries: Pandas, NumPy, SQLAlchemy, Boto3, Requests, Flask, Matplotlib, Seaborn

Databases: PostgreSQL, MySQL, SQLite, schema design, migrations

Cloud & Infrastructure: AWS (EC2, S3, IAM, CloudWatch, SNS), Supabase, Render Cloud

CI/CD & Tools: Git, GitHub Actions/CI-CD, environment variable & secrets management, VS Code, Linux

Data Viz & Serving: Streamlit, Streamlit Cloud, Matplotlib, Seaborn

QA & Testing: Manual, functional, integration & regression testing, UAT, Selenium WebDriver, HP ALM

DATA ENGINEERING PROJECTS

Automated Job Market Intelligence Pipeline — *Python, REST API, Pandas, PostgreSQL, SQLAlchemy, Render, Supabase, CI/CD*

- Built a production-style **ETL pipeline** that ingests live job postings daily from the Adzuna REST API, transforms nested JSON with Pandas, and loads clean structured data into a cloud PostgreSQL database via SQLAlchemy.
- Implemented **serverless cron scheduling** on Render Cloud — compute spins up at midnight, runs the pipeline, and shuts down, eliminating idle hosting cost.
- Applied production security practices: all API keys and connection strings stored in Render's environment variable vault, with zero hardcoded credentials.
- Configured **CI/CD via GitHub → Render**, so pushes to main trigger automatic rebuild and redeployment.
- Engineered transformation logic to parse nested JSON, standardise datatypes (text → ISO timestamps), filter nulls/anomalies, and map fields to a clean relational schema.
- Transferred existing MySQL expertise directly to PostgreSQL (Supabase) — core SQL, joins and schema design carried over, with SQLAlchemy abstracting connection-level differences.

IPL Data Analysis Dashboard — *Python, SQL, SQLite, Pandas, Matplotlib, Seaborn, Streamlit*

- Ingested and processed **260,000+ ball-by-ball IPL records** (2008–2020) into a local SQLite database for structured querying.
- Authored 6 analytical SQL queries covering team performance, top batsmen/bowlers, win-rate by batting/chasing, toss decisions, and player-of-the-match leaderboards.
- Built a 4-chart dashboard (Matplotlib/Seaborn) and deployed an interactive **Streamlit Cloud** app with season filters and KPI cards.

AWS CloudWatch EC2 Monitoring System — *AWS, Python (Boto3), Linux, SNS*

- Provisioned an EC2 Ubuntu instance and configured IAM credentials and AWS CLI access.
- Wrote Boto3 scripts to monitor CloudWatch CPU metrics every 5 minutes and trigger **SNS alerts** above 70% utilisation, confirming a full end-to-end alerting pipeline.

PROFESSIONAL EXPERIENCE

Test Engineer — QA & Application Support, Infosys Ltd (India)

Apr 2022 – Dec 2023

- Delivered end-to-end QA for enterprise CRM and core banking applications across 5+ production releases.

- Performed backend **SQL data validation and migration testing**, identifying and resolving 50+ data discrepancies — directly transferable to data pipeline validation.
- Designed and executed 200+ structured test cases per release using HP ALM, maintaining a zero critical-defect escape rate across 3 consecutive releases.
- Built Selenium WebDriver (Java) regression automation, cutting manual regression effort by ~30% per sprint.
- Owned the defect lifecycle in Jira, collaborating cross-functionally with developers and BAs in Agile teams.

Systems Engineer (Training), Infosys Ltd (India)

Jan 2022 – Mar 2022

- Completed structured training in SDLC, STLC, Agile, Java SE 8 and Selenium WebDriver; certified in Agile Developer and Java SE 8.

Data Science Intern, Innomatics Research Labs (India)

Aug 2021 – Dec 2021

- Completed a part-time data science internship focused on Python programming, data analysis and visualisation using NumPy, Pandas, Matplotlib and Seaborn.
- Built interactive dashboards with **Streamlit** and a web app with **Flask**, gaining early hands-on exposure to full-stack data application development.

EDUCATION

MSc Computing (Merit), Sheffield Hallam University, UK

2024 – 2025

B.Tech Electronics & Communication Engineering, Guru Nanak Institutions Technical Campus, India

2016 – 2020

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- Skills & Best Practices for Cloud Support Associates — Coursera
- Introduction to SQL — SimpliLearn (SkillUp)
- Introduction to Information Technology & AWS Cloud — Coursera
- Cloud Platform Job Simulation — Forage
- Agile Developer Certification — Infosys | Java SE 8 Certification — Infosys

ACHIEVEMENTS

- Best Performer Award — Infosys (DDO Project)
- 3rd Place, Cloud Computing Hackathon — Sheffield Hallam University

WORK AUTHORISATION

Eligible to work in the United Kingdom on the **Graduate Route visa** — no sponsorship required.